Yongjae Shin

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Research Interests

My research aims to make reinforcement learning sample-efficient and reliable for real-world decision making. I study how agents can leverage offline data and generative policies such as flow matching and continually improve them online, with a recent focus on fine-tuning vision-language-action (VLA) models for robot manipulation.

Education

Ph.D., Korea Advanced Institute of Science and Technology (KAIST) 2024 – Present

Smart Information Systems Research Lab (SISReL)
Advisor: Prof. Youngchul Sung

M.S., Korea Advanced Institute of Science and Technology (KAIST) 2022 – 2024

Smart Information Systems Research Lab (SISReL)
Advisor: Prof. Youngchul Sung

B.S. in Electrical Engineering, Korea University 2018 – 2022

Work Experience

Research Intern, LG AI Research Mar 2024 – Dec 2024

Publications: OPT, PARS (ICML 2025); RL-Studio (AAAI 2026 Demonstration).

Publications

* equal contribution, † corresponding author

[9] **Yongjae Shin**, Seongmin Kim, Jongseong Chae, Jongeui Park, Jeongwoo Park, Youngchul Sung. "Steer, Refine, and Select: Adapting Generative Policies for Robot Manipulation via Online RL." *Preprint.*

[8] Minho Park*, Kinam Kim*, Donghu Kim, Byungkun Lee, Dongyoon Hwang, **Yongjae Shin**, Junha Hyung, Hojoon Lee†, Jaegul Choo†. "DiscoDemo: Discovering Efficient and Diverse Robot Demonstrations for Imitation Learning." *Preprint.* [project] [code] [dataset]

[7] **Yongjae Shin**, Jongseong Chae, Seongmin Kim, Jongeui Park, Youngchul Sung. "Adaptive Action Chunking via Multi-Chunk Q Value Estimation." *NeurIPS 2026 Workshop PTA.* [paper]

[6] **Yongjae Shin**, Jongseong Chae, Jongeui Park, Youngchul Sung. "Flow Matching with Injected Noise for Offline-to-Online Reinforcement Learning." *ICLR 2026.* [paper]

[5] Jongseong Chae, Jongeui Park, **Yongjae Shin**, Gyeongmin Kim, Seungyul Han, Youngchul Sung. "Flow Actor-Critic for Offline Reinforcement Learning." *ICLR 2026.* [paper]

[4] Whiyoung Jung, Sunghoon Hong, Deunsol Yoon, Jeonghye Kim, **Yongjae Shin**, Suhyun Jung, Hyundam Yoo, Youngjin Kim, Chanwoo Moon, Woohyung Lim, Soonyoung Lee, Kanghoon Lee. "RL-Studio: A System for

Multi-Phase Reinforcement Learning Experimentation." *AAAI 2026 Demonstration.* [paper]
Work done during an internship at LG AI Research.

[3] **Yongjae Shin**, Jeonghye Kim, Whiyoung Jung, Sunghoon Hong, Deunsol Yoon, Youngsoo Jang, Geon-Hyeong Kim, Jongseong Chae, Youngchul Sung, Kanghoon Lee, Woohyung Lim. "Online Pre-Training for Offline-to-Online Reinforcement Learning." *ICML 2025.* [paper]
Work done during an internship at LG AI Research.

[2] Jeonghye Kim, **Yongjae Shin**, Whiyoung Jung, Sunghoon Hong, Deunsol Yoon, Youngchul Sung, Kanghoon Lee, Woohyung Lim. "Penalizing Infeasible Actions and Reward Scaling in Reinforcement Learning with Offline Data." *ICML 2025.* **Spotlight** [paper]
Work done during an internship at LG AI Research.

[1] Woojun Kim*, **Yongjae Shin***, Jongeui Park, Youngchul Sung. "Sample-efficient and Safe Deep Reinforcement Learning via Reset Deep Ensemble Agents." *NeurIPS 2023.* [paper]

Academic Activities

Conference Reviewer: ICML, NeurIPS, ICLR, AAAI, ICRA

Programming Skills

Python, PyTorch, JAX

Teaching Experience

Teaching Assistant, Probability and Introductory Random Processes, KAIST — Sep 2025 – Dec 2025

Teaching Assistant, Machine Learning Basics and Practices, KAIST — Mar 2025 – Jun 2025